Machine Learning Capstone Project

Predicting house prices, recognizing emotion in text, and grouping districts: regression, classification and clustering

Final Project Report | November 2026

Results at a glance

Task	Model	Main result
Regression	Linear Regression	RMSE 62,562 USD, R^2 = 0.6075
Classification	Logistic Regression	Accuracy 84.88%, Precision 84.36%, Recall 79.06%
Clustering	K-Means (K = 5)	Silhouette Score 0.2816

Code

Google Colab notebook:
https://colab.research.google.com/drive/1QO3u5DIa-m1bJEQIAy9iL9K7HjiItT1A?usp=sharing

The notebook downloads the data and reproduces every number and figure in this report.

I. Introduction

Project Goal

This project walks through the full machine learning pipeline: data acquisition, preprocessing, modeling with three different types of algorithms, evaluation, and reporting. Each algorithm type answers one concrete question:

- **Regression:** predict the median house value of a California district from its location, its housing characteristics, and the income of its residents.
- **Classification:** decide whether a short text describing a personal situation expresses a positive or a negative emotion.
- **Clustering:** group California districts into clusters of similar districts, without using the price as a label.

The classification task connects the project to artificial intelligence and emotion. The EmpatheticDialogues data was created so that conversational AI systems can learn to respond to people's feelings. In this project the model only learns to *recognize* emotion from the words people use. It does not feel anything: it learns statistical patterns between words and emotion labels.

Data Description

Two public datasets are used. Housing data supports the regression and the clustering tasks, and the conversation data supports the classification task.

Dataset	Source	Size	Used for
California Housing	Kaggle (camnugent/california-housing-prices)	20,640 districts, 10 columns	Regression and clustering
EmpatheticDialogues	Hugging Face (Estwld/empathetic_dialogues_llm)	24,850 conversations, 3 columns	Classification

California Housing features. longitude, latitude, housing_median_age, total_rooms, total_bedrooms, population, households, median_income, and ocean_proximity (the only categorical feature). The target is **median_house_value**, a continuous number in US dollars.

EmpatheticDialogues features. The columns are emotion, situation, and first_message. The only input used by the model is the **situation** text, a short description of something that happened to a person. The label comes from the **emotion** column, which has 32 different emotions. These 32 emotions are grouped into one binary target, as explained in the next section.

II. Data Preprocessing & EDA

Summary of Cleaning Steps

California Housing

- **Missing values:** only total_bedrooms had missing values, 207 of them. They were filled with the median of the column, which is not affected by extreme values.
- **Encoding:** ocean_proximity is a nominal feature (no natural order), so it was One-Hot Encoded. One category is dropped to avoid redundancy, leaving four indicator columns.

- **Outliers in the features:** the box plots (Figure 1) show many extreme values in total_rooms, population, households and total_bedrooms. They were kept, because they are real, very large districts and not data errors.
- **Outliers in the target:** the price histogram (Figure 2) has a large spike at 500,001. Every house worth more than this was recorded as 500,001, so the true price of these districts is unknown. The 965 districts at the cap were removed, and 19,675 districts remain.

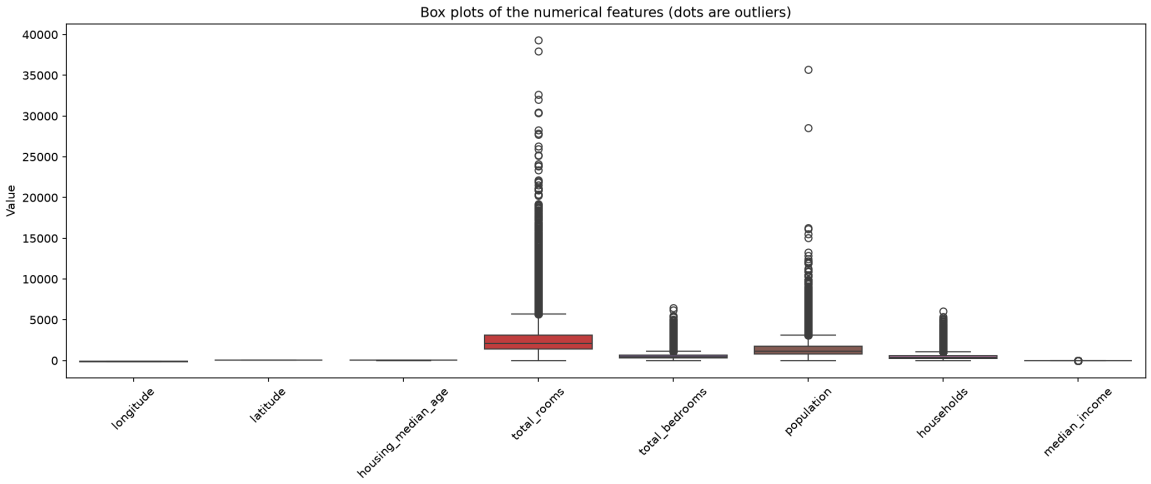


Figure 1. Box plots of the numerical features. Features with small values look flat because all features share one axis. total_rooms, population, households and total_bedrooms have many high outliers.

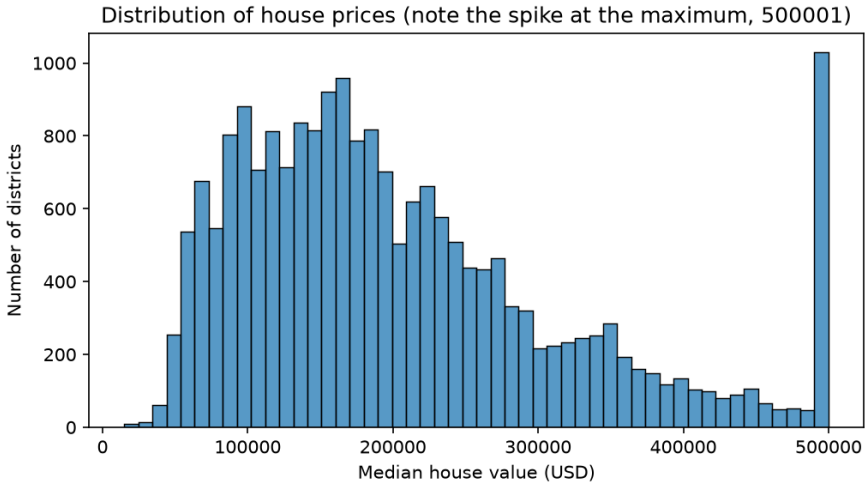


Figure 2. Distribution of house prices. The tall bar at 500,001 is the capped price.

EmpatheticDialogues

- **Missing values:** there are none in the three columns.
- **Building a binary target:** the 32 emotions were grouped into positive (12 emotions, such as excited, proud, grateful, joyful) and negative (16 emotions, such as angry, sad, afraid, guilty). Four emotions that can be either positive or negative depending on the situation (surprised, anticipating, sentimental, nostalgic) were left out. This leaves 21,360 conversations: 12,284 negative (label 0) and 9,076 positive (label 1).
- **Text to numbers:** the situation text was converted with TF-IDF, using the 5,000 most useful words and removing common English stop words.

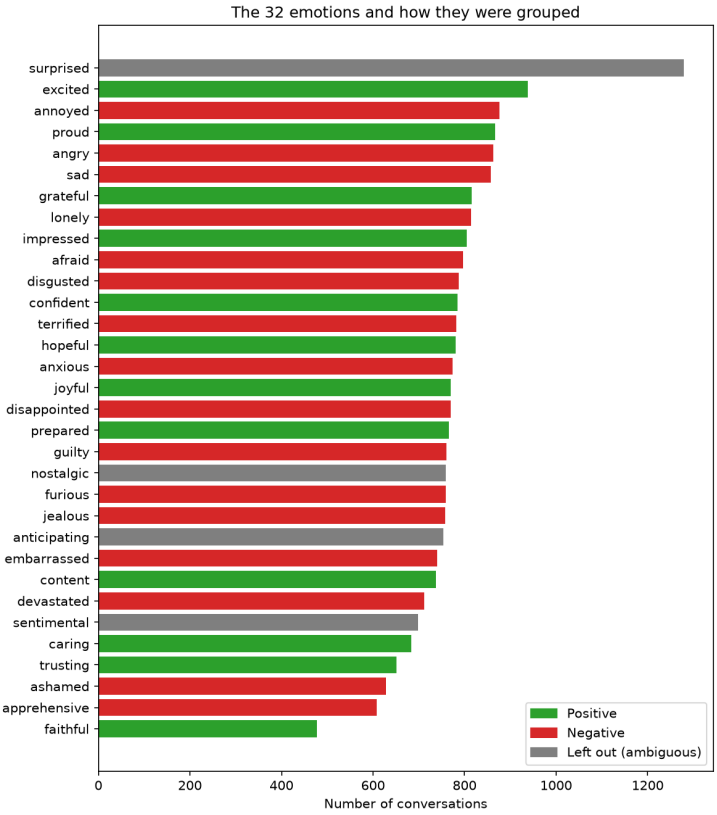


Figure 3. The 32 emotions and how they were grouped. Green is positive, red is negative, gray was left out.

Scaling and Splitting

Both datasets were split into 80% training and 20% test data with a fixed random seed (42), so the results can be reproduced. For the housing data, **StandardScaler** was fitted on the training set only and then applied to the test set, which prevents information from the test data leaking into training. For the text data, TF-IDF already normalizes each row, so StandardScaler was not needed. The classification split is stratified, which keeps the same share of positive and negative texts in both parts. The test sets contain 3,935 districts (housing) and 4,272 texts (emotions).

Key Findings from the EDA

- **Income is the strongest signal.** median_income has a correlation of 0.64 with the house price (Figure 4), far above any other numeric feature.
- **Location matters.** Inland districts tend to be cheaper (correlation -0.50 with the INLAND indicator). The price map (Figure 5) shows the most expensive districts along the coast, around the San Francisco Bay Area and the Los Angeles and San Diego areas.
- **Several features carry almost the same information.** total_rooms, total_bedrooms, population and households are strongly correlated with each other (0.86 to 0.97), and longitude and latitude are correlated at -0.92. This affects how the linear model's coefficients should be read.
- **The emotion classes are slightly unbalanced.** 57.5% of the texts are negative and 42.5% are positive. A model that always answered negative would reach 57.5% accuracy, which is the baseline to beat.
- **Texts are short.** Most situation texts have between 5 and 30 words, and the most common length is about 10 words (Figure 6).

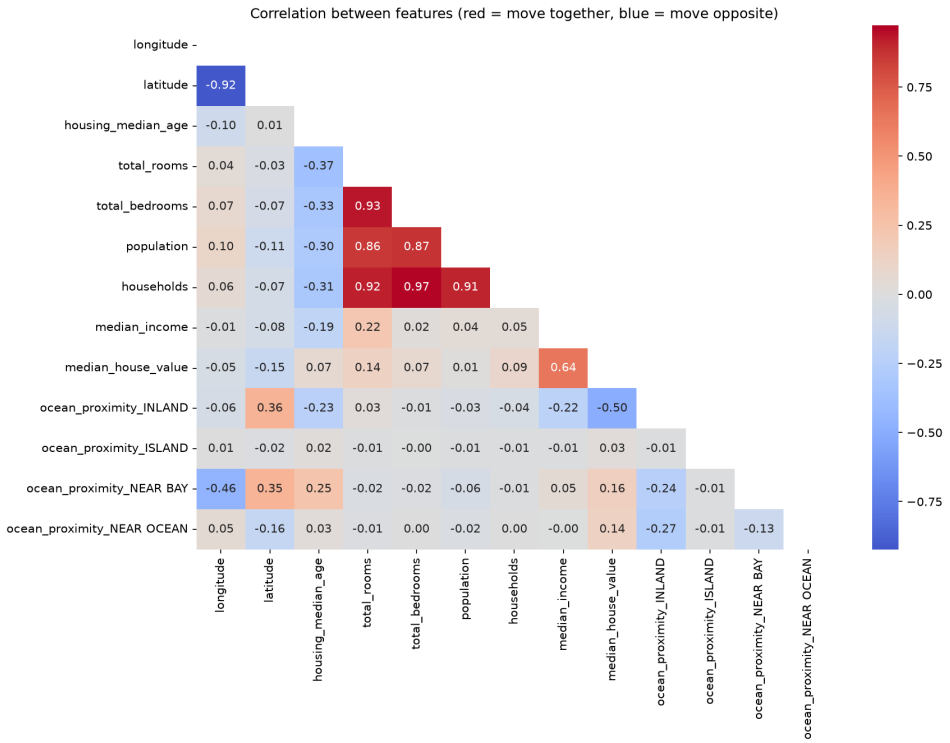


Figure 4. Correlations between the housing features (lower triangle only).

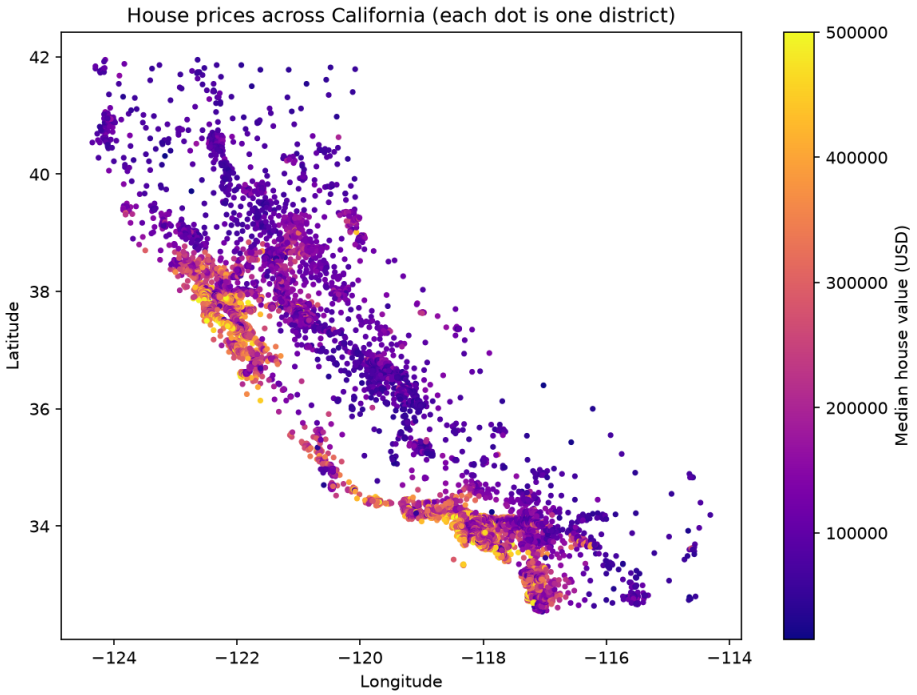


Figure 5. House prices across California. Each dot is one district.

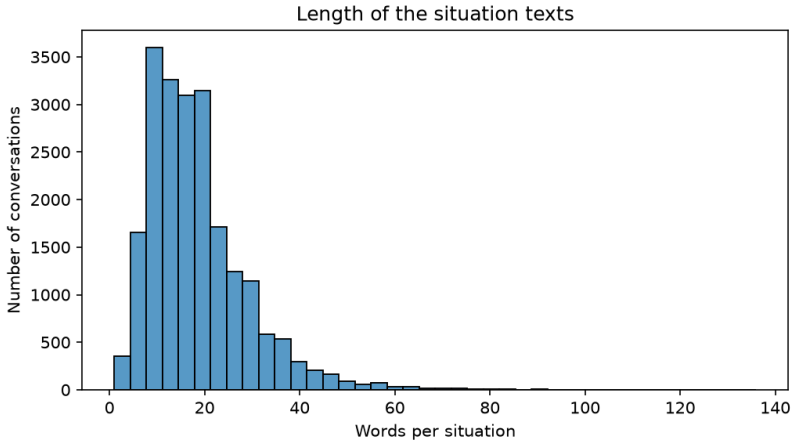


Figure 6. Number of words in each situation text.

III. Modeling and Results

Regression Results: Linear Regression

A Linear Regression model was trained on the scaled housing features to predict median_house_value, and evaluated on the 3,935 test districts.

Metric	Value	Meaning
MSE	3,914,036,126.81	Average squared difference between actual and predicted prices
RMSE	62,562.26 USD	Square root of MSE, in dollars: the typical size of a prediction error
R^2 score	0.6075	Share of the variation in prices explained by the features

Interpretation. An R^2 of 0.6075 means the model explains about 61% of the differences between house prices, and the remaining 39% comes from things the model cannot capture. The RMSE of about 62,600 USD means a typical prediction is off by that amount, which is large when the average price is around 190,000 USD. The model is useful for a rough estimate but not for precise valuation.

Effect of removing the capped prices. Before removing the districts capped at 500,001, the model had an RMSE of 70,060.52 USD and an R^2 of 0.6254. After removal, the error in dollars went down by about 7,500 USD, while R^2 went down slightly. R^2 compares the error with how spread out the prices are, and removing the most expensive districts made the prices less spread out. The two sets of numbers are not directly comparable, because the data and the test set changed. The RMSE, which is in dollars, is the fairer measure here.



Figure 7. Predicted against actual prices. The closer the dots are to the red line, the better the prediction. The model tends to under-predict expensive districts, and one extreme prediction of about -555,000 USD stretches the axes, which shows that a linear model can output impossible negative prices.

What drives the predictions. Figure 8 shows the effect of each feature (all features are standardized, so the bars can be compared). median_income has the largest positive effect, which matches the EDA. Longitude and latitude have large negative effects because prices fall toward the north and the east. Some signs look strange, for example population is negative while households is positive. This happens because these features are strongly correlated with each other, so the model splits their shared effect between them. Individual coefficients of correlated features should therefore not be read as separate causes.

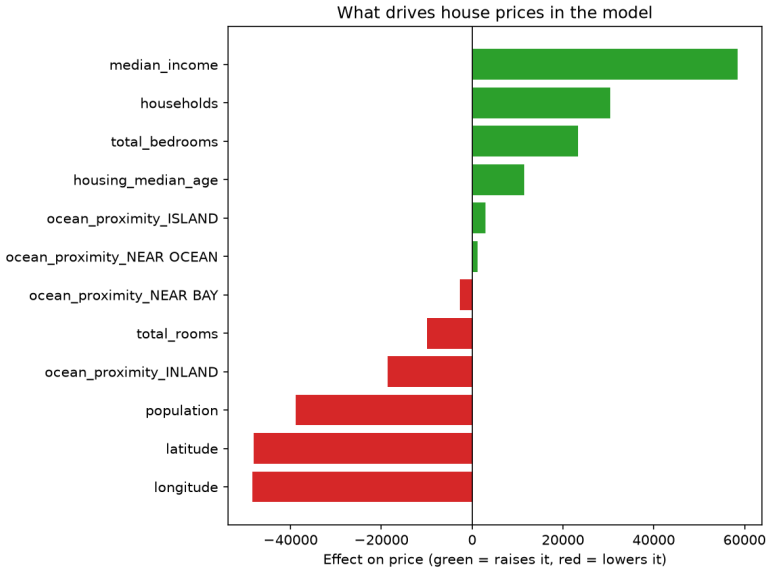


Figure 8. Effect of each feature on the predicted price (green raises it, red lowers it).

Classification Results: Logistic Regression

A Logistic Regression model was trained on the TF-IDF vectors to predict whether a situation text expresses a positive (1) or a negative (0) emotion. It was evaluated on 4,272 test texts.

Metric	Value
Accuracy	0.8488 (84.88%)
Precision (positive class)	0.8436
Recall (positive class)	0.7906

Accuracy. The model classifies 84.88% of the texts correctly. Compared with the 57.5% baseline of always answering negative, this is a large improvement, so the model has learned real patterns in the language.

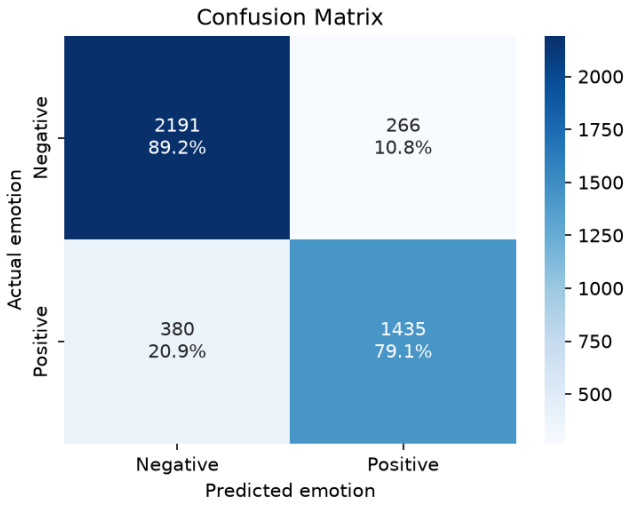


Figure 9. Confusion matrix on the test set. The percentage in each cell is the share of that actual class.

	Predicted negative	Predicted positive
Actual negative	TN = 2,191	FP = 266
Actual positive	FN = 380	TP = 1,435

Confusion matrix analysis. Of the 2,457 negative texts, 2,191 were found correctly (89.2%) and 266 were wrongly called positive (10.8%). Of the 1,815 positive texts, 1,435 were found correctly (79.1%) and 380 were wrongly called negative (20.9%). The model therefore makes more mistakes on positive texts than on negative ones.

Precision and recall trade-off. Precision (84.36%) answers: when the model says positive, how often is it right? Recall (79.06%) answers: of all the truly positive texts, how many did it find? Recall is lower because there are more false negatives (380) than false positives (266): the model plays safe and misses about one positive text in five. Which error matters more depends on the use. If the goal is to notice distress, for example in a support chat, the important class is the negative one, and missing a negative text is the costly mistake. Seen from that side, the model finds 89.2% of the negative texts (recall) and is right 85.2% of the time when it says negative (precision). If this error is too costly, the decision threshold can be lowered to catch more negative texts, at the price of more false alarms.

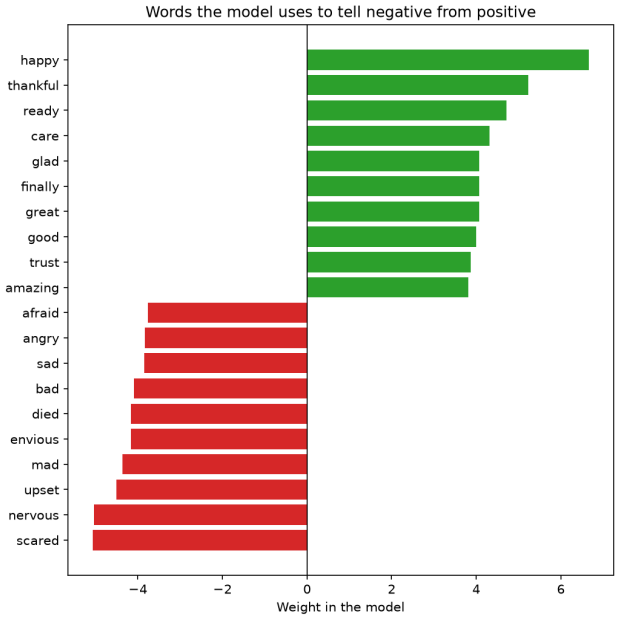


Figure 10. The ten words with the strongest push toward positive (green) and toward negative (red).

What the model learned. The strongest words are sensible: happy, thankful, glad, great and amazing point to positive emotions, while scared, nervous, upset, mad, angry and sad point to negative ones. Words like ready, finally and care also lean positive, which fits emotions such as prepared, hopeful and caring. A limitation is that TF-IDF looks at single words and ignores their order, so a sentence like 'I am not happy' can be misread.

Clustering Results: K-Means

K-Means was run on the housing features without the price, after standardizing them, so the groups are formed from the districts' characteristics only. The Elbow Method plots the inertia (how tight the clusters are) for K from 1 to 10.

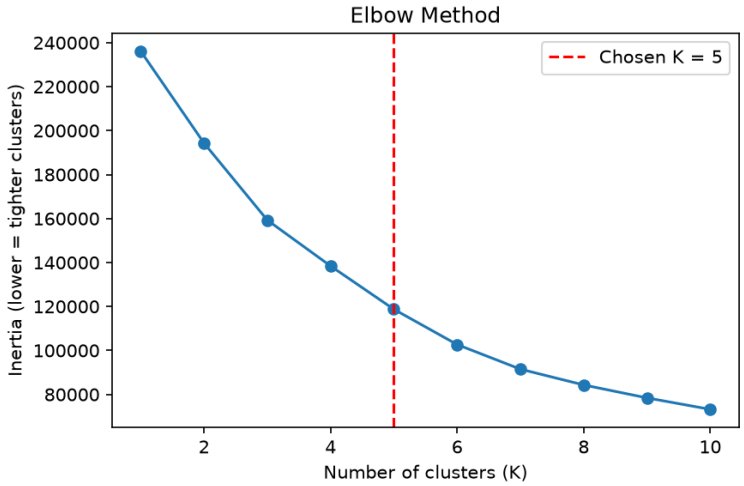


Figure 11. Elbow plot. The red line marks the chosen K = 5.

Choice of K. The curve falls gradually and has no sharp bend, so the elbow is not obvious by eye. K = 5 was selected with a simple rule: the point that lies farthest from the straight line joining the first and the last point of the curve.

Silhouette Score. The final score is 0.2816. Scores range from -1 to 1, and a value this low means the clusters are present but overlap noticeably. This is expected for housing data, which varies smoothly across the state and has no clear natural groups.

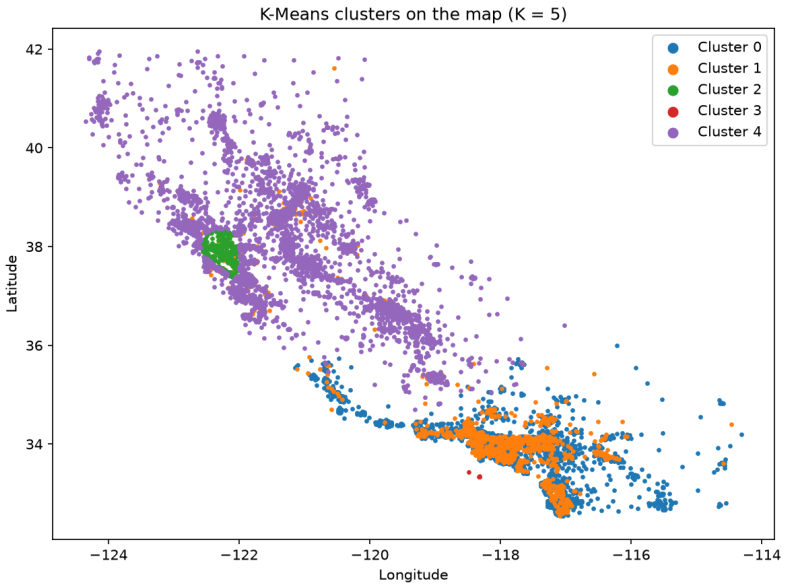


Figure 12. The five clusters on the map of California.

Interpretation of the clusters. The table shows the average of key features in each cluster, and the map shows where the clusters are. Because latitude and longitude are among the features, the clusters follow geography.

Cluster	Districts	Avg. income	Avg. price (USD)	Avg. house age	Avg. population
0	9,861	3.7	204,309	29.6	1,299
1	1,473	4.1	207,462	15.9	4,126
2	2,018	3.9	236,435	38.0	1,131
3	5	2.7	380,440	42.4	668
4	6,318	3.4	156,330	26.4	1,136

- **Cluster 4** covers northern and central California and has the lowest average price, about 156,000 USD.
- **Cluster 0** is the largest group and lies mostly in Southern California. It is a middle group with average values for everything.
- **Cluster 1** sits mostly in the Los Angeles basin. These are the newest and most crowded districts, with an average of 4,126 residents and houses that are 16 years old on average.
- **Cluster 2** is the core of the San Francisco Bay Area. It has the oldest housing (38 years on average) and higher prices, about 236,000 USD.
- **Cluster 3** has only 5 districts, isolated points off the southern coast. The average price is high (about 380,000 USD), which fits the ocean_proximity category ISLAND.

IV. Conclusion

Best-Performing Model for Each Task

Task	Model	Result	Verdict
Regression	Linear Regression	RMSE 62,562 USD, R^2 0.6075	Reasonable baseline; explains about 61% of price variation
Classification	Logistic Regression	Accuracy 84.88%	Strong, well above the 57.5% baseline
Clustering	K-Means, K = 5	Silhouette 0.2816	Geographically meaningful but overlapping clusters

The classification model performed best relative to its baseline: with plain word counts it reaches almost 85% accuracy. The regression model is a reasonable first model, and the clustering found structure that follows the geography of the state.

Challenges Encountered

- **Capped prices.** The 500,001 cap hides the real price of the most expensive districts, so these rows had to be removed.
- **Strongly correlated features.** Rooms, bedrooms, population and households carry almost the same information, which makes the individual coefficients of the linear model hard to interpret.
- **Turning emotions into a binary target.** Grouping 32 emotions into positive and negative required judgment, and four ambiguous emotions had to be left out.
- **Text as input.** Text had to be converted to numbers with TF-IDF, which ignores word order and negation.
- **No clear elbow.** Choosing K was difficult because the inertia curve has no sharp bend, and the final Silhouette Score is low.

Potential Next Steps

- **Gradient Boosting** for regression, to capture non-linear relationships between income, location and price, and to avoid negative predictions.
- **PCA** before K-Means, to remove the redundancy between the correlated features and possibly improve the Silhouette Score.
- **Richer text models** such as word embeddings or transformer models, which understand word order and negation, and a multi-class classifier that predicts all 32 emotions.
- **Threshold tuning** for the classifier, to trade precision for recall depending on which mistake is more costly.
- **Feature engineering** for the housing data, for example rooms per household, to reduce the overlap between correlated features.

Reproducibility

Code (Google Colab): https://colab.research.google.com/drive/1QO3u5DIa-m1bJEQIAy9iL9K7HjiItT1A?usp=sharing. Public GitHub repository: ML-Capstone-Project. A fixed random seed (42) makes every number reproducible.